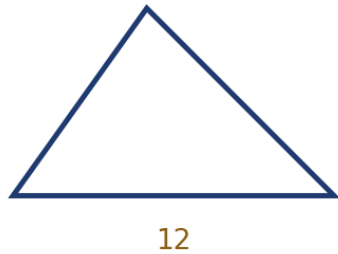


Geometry & Trigonometry

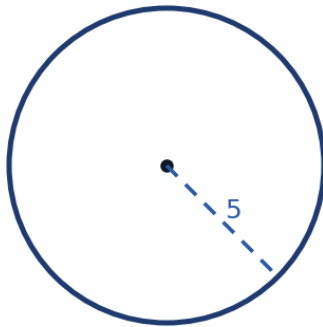


Area and perimeter of basic shapes

- 1 Find the area of a triangle with base 12 and height 7.

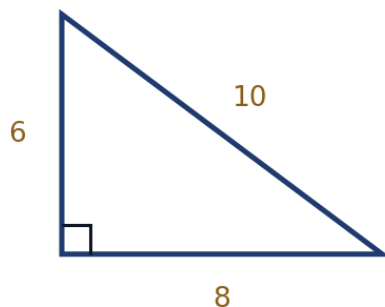


- 2 Find the area and circumference of a circle with radius 5, in terms of π .

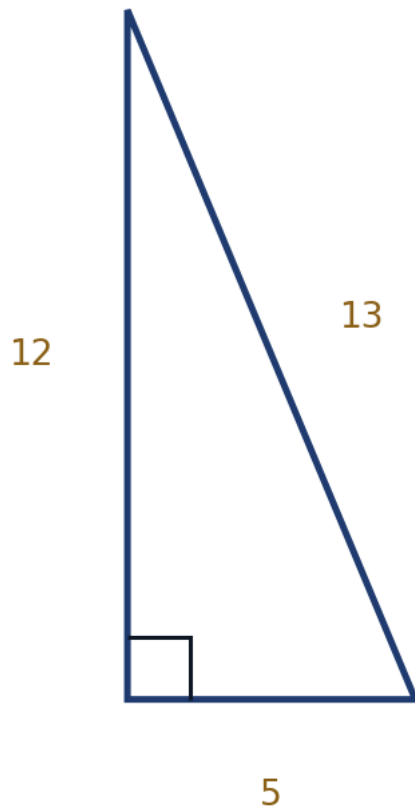


The Pythagorean theorem

- 3 A right triangle has legs 6 and 8. Find the length of the hypotenuse.

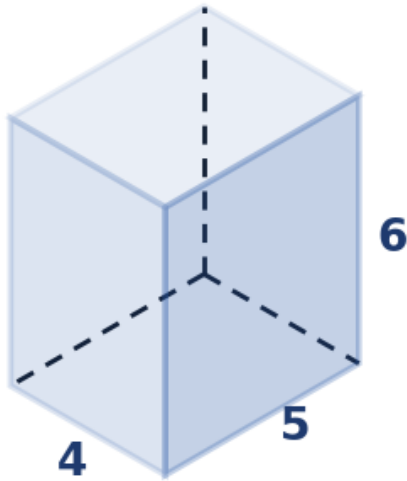


- 4 A ladder 13 feet long leans against a wall, with its base 5 feet from the wall. How high up the wall does the ladder reach?

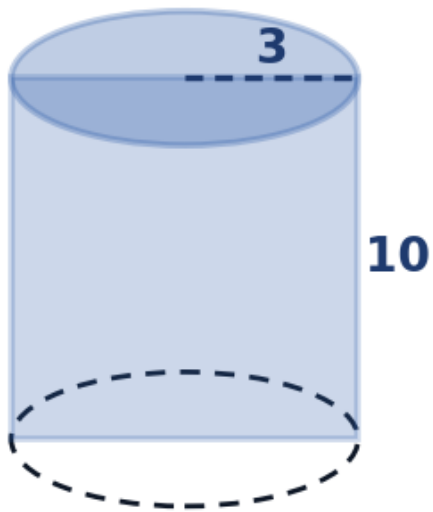


Volume and surface area of solids

- 5 Find the volume of a rectangular prism with dimensions 4 by 5 by 6.



- 6 Find the volume of a cylinder with radius 3 and height 10, in terms of π .

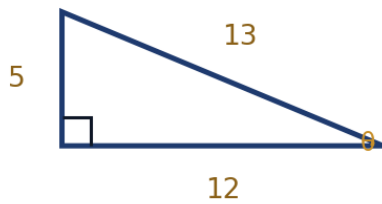


Similar triangles and proportional reasoning

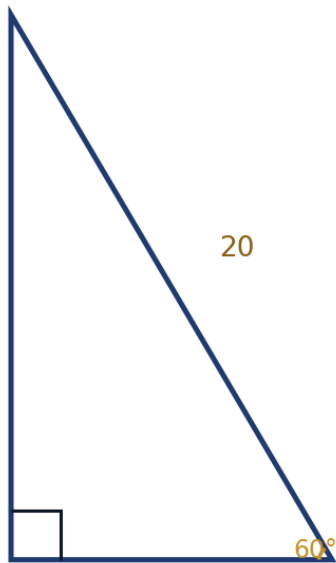
- 7 Triangle ABC is similar to triangle DEF , with $AB = 6$, $DE = 9$, and $BC = 8$. Find EF .
- 8 A tree casts a shadow 20 feet long, while a 6-foot person standing nearby casts a shadow 4 feet long. Find the height of the tree using similar triangles.
-

Right triangle trigonometry

- 9 In a right triangle, the angle θ has an opposite side of length 5 and hypotenuse 13. Find $\sin \theta$, $\cos \theta$, and $\tan \theta$.

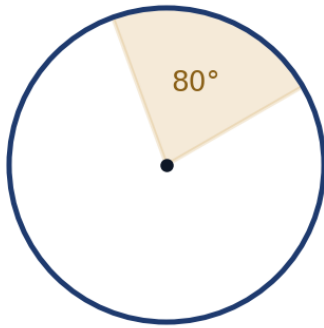


- 10 A 20-foot ladder leans against a wall, making a 60° angle with the ground. Find the height the ladder reaches on the wall, to the nearest tenth.



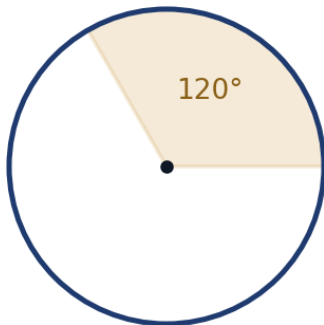
Angle relationships and circle theorems

- 11** Two parallel lines are cut by a transversal. If one angle measures 65° , find the measure of its corresponding angle and its co-interior (same-side interior) angle.
- 12** A central angle in a circle measures 80° . Find the measure of the inscribed angle that subtends the same arc.

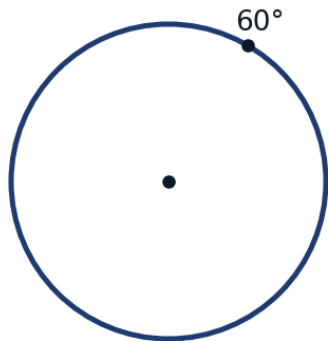


The unit circle and radian measure

- 13** Convert 120° to radians, in terms of π .

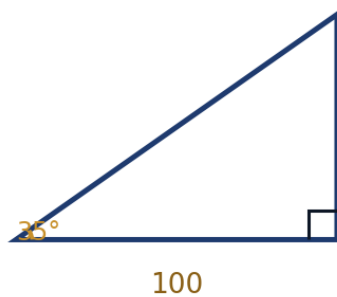


- 14 Find the exact value of $\cos\left(\frac{\pi}{3}\right)$ using the unit circle.



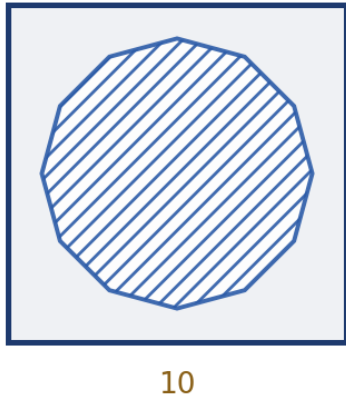
Applying geometry and trigonometry to real-world contexts

- 15 A surveyor stands 100 feet from the base of a building and measures the angle of elevation to the top as 35° . Find the height of the building, to the nearest foot.

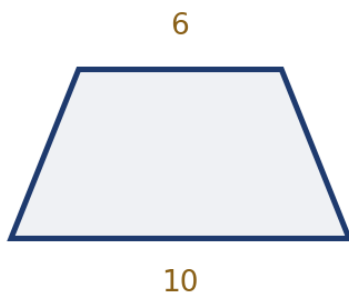


Composite area and shaded regions

- 16 A square with side length 10 has a circle of radius 4 cut out of its center. Find the remaining area, in terms of π .

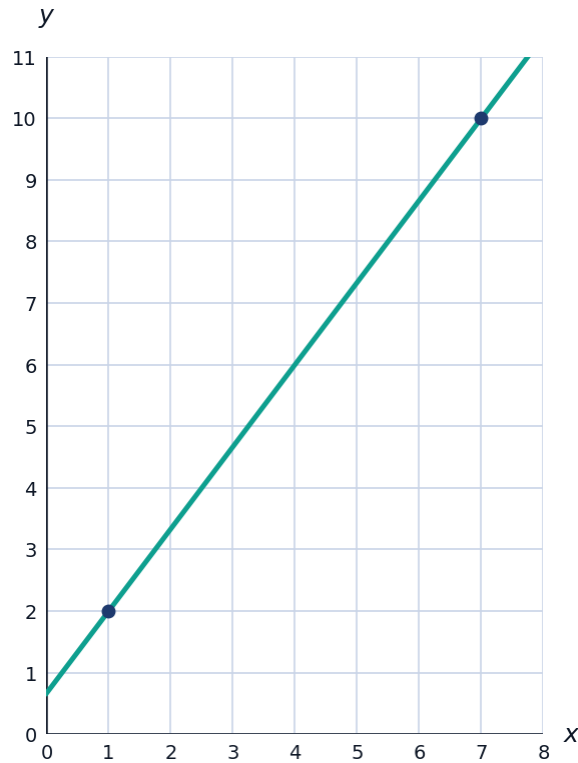


- 17 Find the area of a trapezoid with parallel sides 6 and 10, and height 5.

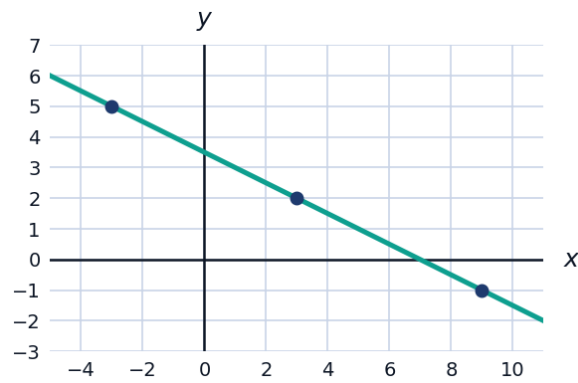


Coordinate geometry: distance and midpoint

- 18 Find the distance between the points $(1, 2)$ and $(7, 10)$.

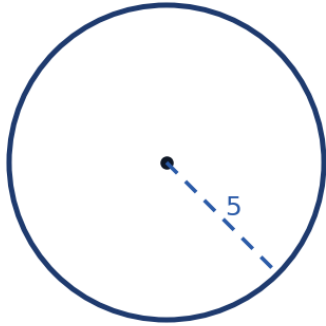


- 19 Find the midpoint of the segment connecting $(-3, 5)$ and $(9, -1)$.

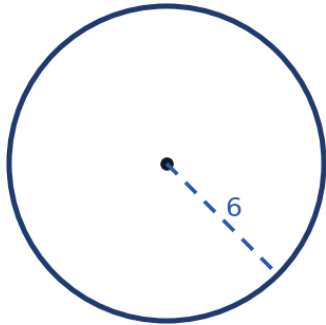


Equation of a circle

- 20 Write the equation of a circle with center $(2, -3)$ and radius 5.

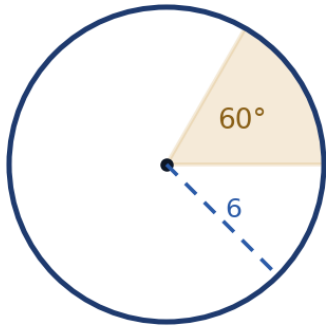


- 21 Find the center and radius of the circle $(x + 1)^2 + (y - 4)^2 = 36$.



Arc length and sector area

- 22 A circle has radius 6. Find the arc length of a 60° sector, in terms of π .



- 23 Find the area of a 90° sector of a circle with radius 8, in terms of π .

